

Nagar Drishti

AI-powered urban intelligence and municipal command platform for road-defect detection, fleet monitoring, ticket routing, and cross-agency governance.

Prepared September 10, 2026 | Repository: remix-nagar-drishti

Project Reference

The application combines a React/Vite command-center interface, an Express REST and WebSocket backend, and an edge-to-cloud computer-vision pipeline documented in the repository.

AI Studio project https://ai.studio/apps/64a79b0a-395a-4019-8106-3e1cf2baa1af
Architecture documentation https://github.com/ Repository file: ai_engine/README.md
Local setup and execution https://github.com/ Repository file: README.md

System Research Summary

- **Edge sensing:** synchronized camera streams on NVIDIA Jetson hardware run object detection, tracking, and OCR locally.
- **Telemetry:** event-driven JSON and WebP/JPEG evidence is transmitted over MQTT or WebSocket rather than streaming raw video.
- **Cloud intelligence:** Haversine-distance DBSCAN groups repeated detections into canonical defect clusters.
- **Decision support:** traffic analytics and Gemini structured function calling help route validated incidents into municipal work orders.
- **Command center:** dashboards expose live detections, maps, fleet activity, analytics, alerts, and cross-agency workflows.

Dataset and Evaluation Basis

3,184 total images	6,463 annotated road-damage boxes
2,231 training images / 4,529 boxes	474 + 479 validation and test images

The cleaned dataset retains one class, **RoadDamages**, remapped from the original class index to class 0. Other source categories were removed for this model configuration.

Attribution gap: the repository records the cleaned dataset statistics but does not currently identify the original public dataset source, license, or citation. Those details should be added before external publication.

Technical and Research References

Ultralytics YOLO

<https://docs.ultralytics.com/>

Object detection and segmentation model documentation.

ByteTrack

<https://arxiv.org/abs/2110.06864>

Multi-object tracking by associating every detection box.

DBSCAN in scikit-learn

<https://scikit-learn.org/stable/modules/clustering.html#dbscan>

Density-based clustering reference used for spatial deduplication.

Gemini function calling

<https://ai.google.dev/gemini-api/docs/function-calling>

Structured model interaction for dispatch and work-order generation.

NVIDIA Jetson Orin Nano

<https://developer.nvidia.com/embedded/jetson-orin-nano-developer-kit>

Target hardware family for edge inference.

MQTT

<https://mqtt.org/>

Lightweight publish-subscribe telemetry protocol.

OpenStreetMap

<https://www.openstreetmap.org/>

Open mapping reference for map-based civic operations.

Government of India Open Data

<https://data.gov.in/>

Potential source for public civic and transport datasets.

Local Deliverables

- **ai_engine/README.md** - edge-to-cloud architecture and execution notes.
- **ai_engine/dataset_summary.json** - cleaned dataset counts and class definition.
- **README.md** - frontend/backend setup and test commands.

Nagar Drishti | Reference and Research Brief | This document distinguishes repository-documented implementation details from external technical references.